

UNIT 7 • HOMEWORK PACKETS

Answer Key

Grade 6 Mathematics • Unit 7: Integers and Coordinate Plane • Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 7-1*Integers and Opposites • Integers in Real Situations***Sheet 1 • Integers and Opposites**

1. a. -12 b. 40 c. -9 d. 15 e. -25 f. 7
2. Points at -6 and 6; each is 6 units from 0.
3. A: -7, opposite 7; B: -2, opposite 2; C: 3, opposite -3; D: 9, opposite -9
4. Opposites are on different sides of 0, so the sign changes. -15 is 15 units left of 0; its opposite is 15 units right of 0. The opposite of -15 is 15.

Sheet 2 • Integers in Real Situations

1. a. -35 b. 20 c. -4 d. -8
2. a. 9 b. -15 c. -4
3. A at -3, B at 3, C at -3. A and C are the same point: the opposite of the opposite of -3 is -3.
4. a. noon 6, midnight -6 b. They are opposites: each is 6 units from 0, on different sides.
5. No. The opposite of a negative number is positive. Example: the opposite of -7 is 7, which is positive.

TEST PRACTICE

6. B
7. B

LESSON 7-2*Rational Numbers on a Number Line • Opposites of Rational Numbers***Sheet 1 • Rational Numbers on a Number Line**

1. A = -1.75 (-1 3/4) B = -0.5 (-1/2) C = 1.25 (1 1/4)
2. -2.5 halfway between -3 and -2; 1/2 halfway between 0 and 1; -0.75 three fourths left of 0; 1 1/4 one fourth right of 1.
3. a. -5 and -4 b. 3 and 4 c. -1 and 0 d. -3 and -2
4. a. -3.50 b. 2 1/2 c. -4.5
5. a. frog 1.5, fish -2.25 b. the frog (1.5 units from 0 is less than 2.25 units from 0)
6. -1.5 is 1.5 units left of 0, so it is past -1. It belongs halfway between -2 and -1.

Sheet 2 • Opposites of Rational Numbers

1. a. -4.8 b. 2/3 c. 0.05 d. 0 e. 1.6 f. 5/8
2. Q at 2.25 (2 1/4), one fourth to the right of 2.
3. 12.5 points; it is 12.5 points from 0 (same distance as -12.5)
4. a. M = -3.75 (-3 3/4) b. N = 3.75 (3 3/4)
5. -(-2/5) means the opposite of -2/5. The opposite of a negative number is positive, so -(-2/5) = 2/5.

TEST PRACTICE

6. B
7. C

LESSON 7-3*Absolute Value as Distance • Absolute Value as Total Distance***Sheet 1 • Absolute Value as Distance**

1. a. 14 b. 8 c. 0 d. 2.5 e. 3/4 f. 1 1/2
2. a. $|A| = |-7| = 7$ b. $|B| = |5| = 5$ c. point A
3. a. $>$ ($9 > 7$) b. $=$ ($4 = 4$) c. $<$ ($-10 < 3$) d. $=$ ($6.5 = 6.5$) e. $<$ ($12 < 15$) f. $<$ ($0 < 1$)
4. a. $|-35| = 35$ and $|20| = 20$ b. Rosa's ($35 > 20$)
5. -2, -1, 0, 1, 2 (each is less than 3 units from 0)
6. 4.5 and -4.5. Both are 4.5 units from 0, on different sides.

Sheet 2 • Absolute Value as Total Distance

1. a. $5 + 6 = 11$ b. $1 + 6 = 7$ c. $5 - 1 = 4$
2. a. 10 b. 8 c. 6.5
3. $8 + 15 = 23$ degrees
4. -3 and 9 are on opposite sides of 0: $3 + 9 = 12$ floors
5. -9 and 5 are on opposite sides of 0, so he should add: $9 + 5 = 14$ units.

TEST PRACTICE

6. B
7. B

LESSON 7-4*Comparing Numbers on a Number Line • Ordering Integers and Rational Numbers***Sheet 1 • Comparing Numbers on a Number Line**

1. J = -4.5, K = -1, L = 3. a. J < K b. L > K c. J < L
2. a. < b. > c. < d. > e. > f. >
3. Points at -3.5 and -0.5; $-3.5 < -0.5$

Sheet 2 • Ordering Integers and Rational Numbers

1. -3.25, -2.5, -3/4, 1.5, 2 1/4
2. a. -9, -6, 0, 4 b. -3, -0.3, -0.03
3. a. -2 b. -1.04 c. -1/2 d. 0

4. a. $-5 < -3$ (or $-3 > -5$) b. Malik (-5 is the lower score)
5. Monday was colder: $-7 < -2$ (or $-2 > -7$), because -7 is farther left on the number line.
6. Yes. $|-6| = 6$, and -6 is left of 6 on the number line, so $-6 < 6$.
4. shark (-45.5), dolphin (-30.25), turtle (-12), jellyfish (-8.75)
5. He ordered by distance from 0, as if the numbers were positive, so his list goes greatest to least. For negatives, farther from 0 is less. Correct order: -0.6 , $-1/2$, -0.4 .

TEST PRACTICE

6. B
7. B

LESSON 7-5

Plotting Points on the Coordinate Plane · Reflecting Points Across Axes

Sheet 1 · Plotting Points on the Coordinate Plane

1. A (4, 3) I; B (-4, 2) II; C (-2, -4) III; D (3, -4) IV
2. P(2, 4) in I; Q(-4, -1) in III; S(-3.5, 3) in II; G(4, -2.5) in IV. a. the pool b. the garden
3. a. Quadrant III b. Quadrant IV c. on the y-axis d. Quadrant II e. on the x-axis f. Quadrant IV
4. No. Order matters. (-3, 5) is 3 left and 5 up (Quadrant II). (5, -3) is 5 right and 3 down (Quadrant IV).

Sheet 2 · Reflecting Points Across Axes

1. a. B = (-4, -3) b. C = (4, 3)
2. a. (6, -2) b. (4, -7)
3. (7, 2.5)
4. K = (-6, 5). Reflecting across the x-axis flips the sign of y, so flip it back: -5 becomes 5.
5. He changed the y-coordinate. Across the y-axis, y stays the same and x changes sign. Correct point: (3, -8).

TEST PRACTICE

6. B
7. A

LESSON 7-6

Distance Between Points on a Grid · Comparing Distances on the Plane

Sheet 1 · Distance Between Points on a Grid

1. A(-4, 3), B(3, 3), C(3, -2), D(-4, -4). a. $4 + 3 = 7$ units b. $3 + 2 = 5$ units c. $3 + 4 = 7$ units
2. a. 8 b. 5 c. 9 d. 11 e. 5 f. 6.5 g. 10 h. 4.5
3. Same x; $3 + 5 = 8$ blocks
4. Q = (-4, -3)
5. Both y-coordinates are negative (same side of the x-axis), so subtract: $6 - 2 = 4$ units.

Sheet 2 · Comparing Distances on the Plane

1. H(-4, 2), L(3, 2), P(-4, -3). a. $4 + 3 = 7$ blocks b. $2 + 3 = 5$ blocks c. the Park, by 2 blocks
2. a. $8 - 3 = 5$ b. $6 + 7 = 13$ c. $7 - 1 = 6$ d. $2.5 + 4.5 = 7$
3. Ana: $5 + 6 = 11$ units. Jay: $4 + 6 = 10$ units. Ana runs 1 unit farther.
4. Add. The y-coordinates 7 and -2 have different signs, so the path crosses the x-axis: $7 + 2 = 9$ units.

TEST PRACTICE

5. A - JK = $4 + 6 = 10$, MN = $3 + 5 = 8$
6. C

LESSON 7-7

Polygons and Perimeter on the Plane · Designing Polygons with Scale

Sheet 1 · Polygons and Perimeter on the Plane

1. A(-4, 3), B(2, 3), C(2, -2), D(-4, -2). a. $4 + 2 = 6$ units b. $3 + 2 = 5$ units c. $6 + 5 + 6 + 5 = 22$ units d. $6 \times 5 = 30$ square units
2. a. $3 + 4 = 7$ units b. $2 + 4 = 6$ units c. $\frac{1}{2} \times 7 \times 6 = 21$ square units
3. Sides $5 + 6 = 11$ m and $3 + 4 = 7$ m; P = $11 + 7 + 11 + 7 = 36$ m
4. Each side is $3 + 2 = 5$ units; P = $4 \times 5 = 20$ units
5. (-2, -1). It must line up with (-2, 5) vertically ($x = -2$) and with (3, -1) horizontally ($y = -1$).

Sheet 2 · Designing Polygons with Scale

1. A(-3, 4), B(-3, -2). a. $4 + 2 = 6$ units b. $6 \times 5 = 30$ ft c. $1,200 \div 30 = 40$ ft d. $40 \div 5 = 8$ units
2. a. 28 m b. 9 units
3. $14 \text{ ft} \times 10 \text{ ft} = 140 \text{ ft}^2$
4. Grid sides $2 + 2 = 4$ units (12 ft) and $4 + 1 = 5$ units (15 ft). P = $12 + 15 + 12 + 15 = 54$ ft
5. Lily subtracted $5 - 4$. The signs differ, so add: $4 + 5 = 9$ units; $9 \times 3 = 27$ ft.

TEST PRACTICE

6. B
7. A